An Interactive Exploration of OECD Well-Being Trends (2004-2024)

Website: https://com-480-data-visualization.github.io/betterlife/

1. Project Goal

This project aims to create an interactive web-based visualization exploring the OECD Better Life Index dataset, covering the period from 2004 to 2024. The primary goal is to allow users to visually investigate how overall well-being and its constituent dimensions (e.g., Housing, Income, Jobs, Community, Education, Environment, Civic Engagement, Health, Life Satisfaction, Safety, Work-Life Balance) have evolved over time across OECD member countries and selected non-OECD partner countries.

This project distinguishes itself from existing visualizations, such as the static OECD "Better Life Index" flower visualization, in several key ways:

- **Temporal Dynamics:** Unlike static snapshots, our core focus is on the evolution of well-being metrics over two decades (2004-2024) via an interactive time slider.
- **Data Representation**: While some visualizations might map raw survey data points, our visualization will focus on presenting the aggregated/averaged index scores provided in the dataset, facilitating clearer country-level comparisons and trend analysis over time.

2. Core Visualization Concept (Minimal Viable Product - MVP)

The central piece of the MVP will be an interactive choropleth map of the world, focusing on OECD and partner countries. Below are the individual pieces to implement:

- **Map Display**: The map will display countries colored according to the selected well-being dimension score for a specific year.
- **Color Scale:** An appropriate sequential color scale will be used to show how countries on the map performed on that chosen dimension, indicated by a legend.
- **Graying out of countries**: Countries with no data for the selected dimension should automatically be grayed out. There should also be an option for users to toggle on and off to hide the countries completely from the map. For OECD countries with only partial data for some dimensions, this mechanism should make them "re-appear" on the map when going from a dimension with no available data, to one with data.
- **Time Slider:** A slider will allow users to select a year between 2004 and 2024. Adjusting the slider will dynamically update the map colors and displayed data based on the selected year, with smooth transitions.
- **Dimension list:** A drop-down menu or radio button list will let users select their dimension of choice for the visualization to use.
- **Interactivity:** Hovering over a country on the map will trigger a tooltip displaying the country's name and its specific score for the selected dimension for that year.
- **Basic Information Panel:** Clicking on a country would zoom into the region, and hide all the other countries so the user can focus.

3. Tools we (think) we will use

For the Interactive Geographical Map:

- *d3-geo* – To create geographical projections and generate SVG paths from GeoJSON/ TopoJSON data.
- *d3-selection* – For selecting and updating DOM elements (e.g., highlighting countries).
- *d3-scale* – To define color scales for visualizing well-being scores.
- *d3-array* – To sort, group, and filter country data.
- *d3-topojson* – To work with efficient TopoJSON geometries for mapping.
- *d3-transition* – To animate transitions between years smoothly.
- *d3-zoom* – To handle zoom in and out of the map.

For the Time Slider (2004–2024):

- *GeoJSON* – Format to encode country boundaries and properties (e.g., ISO codes, names).
- *TopoJSON* – An efficient extension of GeoJSON for shared geometries; reduces file size and improves performance.
- *GDAL* (for preprocessing if needed) – Command-line tool to convert or clip geospatial data (e.g., to match OECD countries only).

Lectures we will need:

- `Lecture on Perception and Color` – To design effective color scales using visual perception principles, avoid misleading color maps, and support accessibility (e.g., colorblind users).
- `Lecture on Maps` – Crucial for understanding GeoJSON/TopoJSON structures, map projections, and D3's geographic modules.
- `Lecture on Graphs` – For structuring and optionally visualizing relationships between well-being dimensions (e.g., co-variation across dimensions or country similarity).
- `Lecture on Storytelling` – To design a compelling and intuitive user experience that guides exploration and highlights key insights through dynamic narrative techniques.

4. Potential future improvements

- *d3-brush* – To allow users to select a range of years for deeper analysis. Geospatial and Mapping Tools (from lecture)
- *Leaflet.js* – A popular JS library for interactive tile-based maps. Could be used in combination with D3 for smoother zoom/pan behavior.
- *Turf.js* – For more advanced spatial calculations, like centroids or region boundaries.

Note that the provided sketch on the next page consolidates multiple interactive states or potential views for illustrative purposes, even if they wouldn't necessarily appear simultaneously or in that exact configuration in the final interactive prototype.

The final implementation will likely involve dynamic changes to the view based on user interaction, which is difficult to fully capture in static sketches.